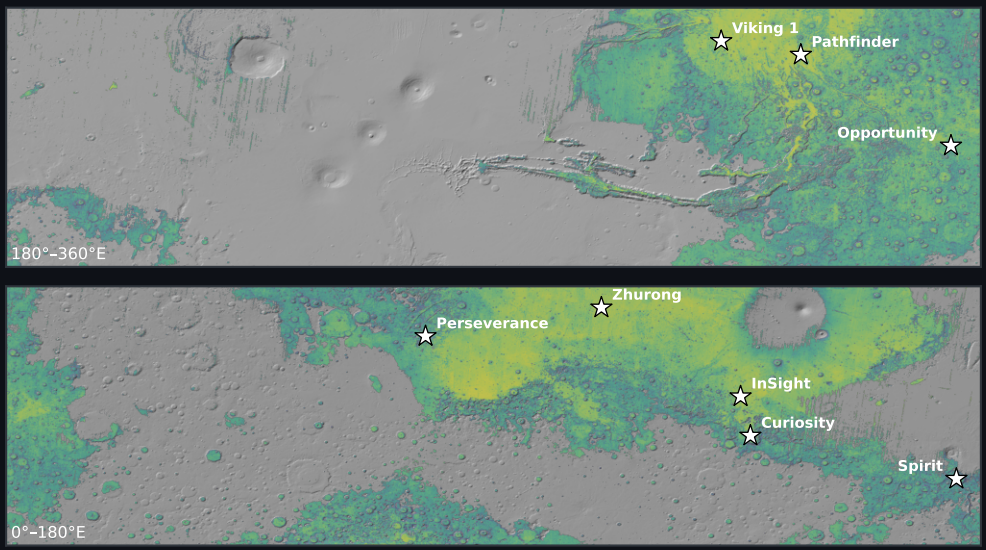
Can Earth GIS methods pick a landing site on Mars?

Same multi-criteria workflow used for site selection on Earth, applied to public NASA/ESA orbital data for Mars' ±30° band.



Result: 8 of 8 real landing sites scored in the top 20 %.

Six criteria, three free datasets

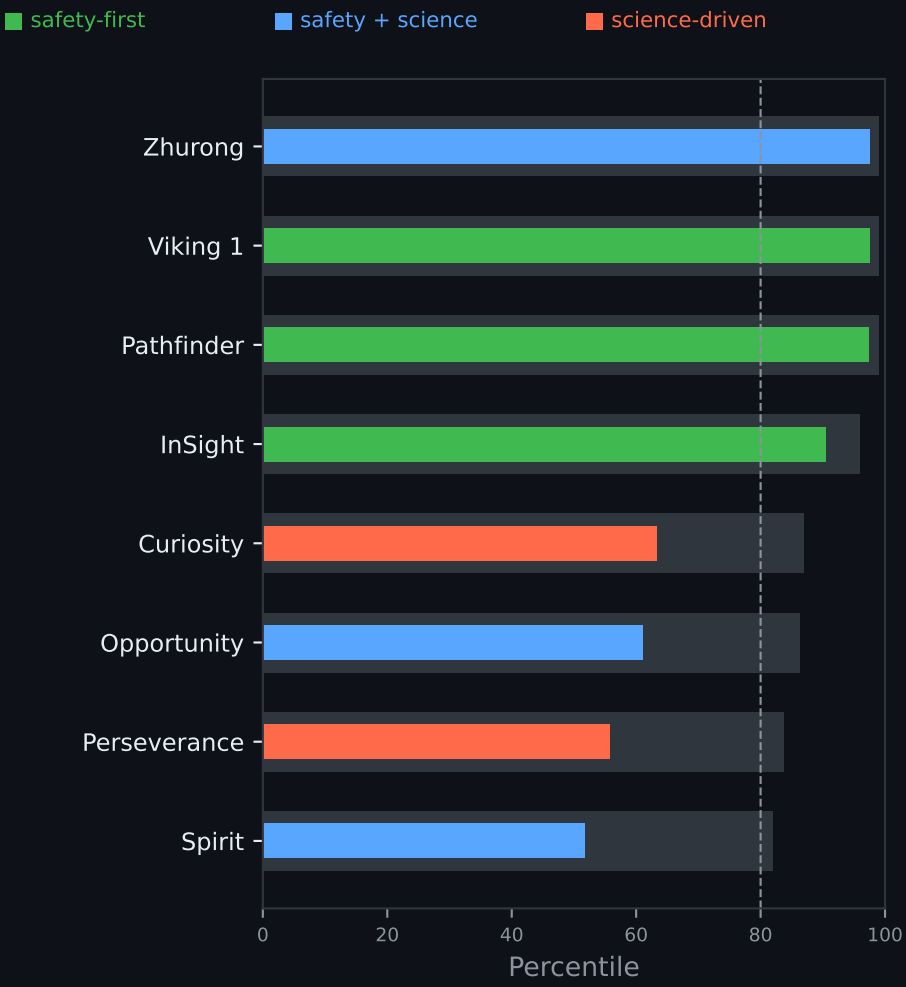
Everything on one Mars equirectangular grid (IAU_2015:49910), 1 km pixels, ±30° latitude.

Criterion	Source	AHP weight
Slope	MOLA–HRSC DEM	**27 %**
Roughness	MOLA–HRSC DEM	**27 %**
Elevation	MOLA–HRSC DEM	**15 %**
Thermal inertia	TES (Putzig & Mellon)	**15 %**
Crater density	Robbins & Hynek, 384k craters	**9 %**
Latitude	grid	**6 %**

Hard constraints remove ground above +1 km, slopes over 10° and dust (thermal inertia < 100). AHP consistency ratio = 0.007 (limit 0.10).

Validation: 8 real landing sites

Mean score within 10 km, ranked against 1,000 random points.
Grey: vs. the whole band · colour: vs. ground that is already safe.



The misses are the interesting part

Even against ground that is already safe, Zhurong, Viking 1, Pathfinder and InSight stay in the top 10 %.

Jezero (Perseverance) and Gale (Curiosity) drop to the 56th and 63th percentile.

They were chosen for science, and needed precision landing (guided entry, Terrain-Relative Navigation) to get there.

The model encodes engineering safety only; the gap measures what science bought at the cost of margin.

90–98

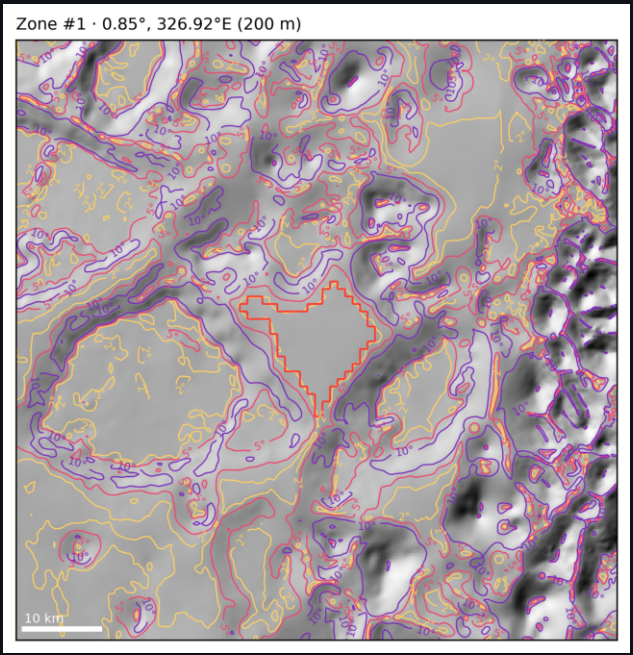
percentile
Viking 1, Pathfinder, InSight, Zhurong

56–63

percentile
Jezero & Gale

Best candidate & robustness

Zone #1: 0.85° N, 326.92° E · 138 km² · Ares Vallis floor



- Checked at 200 m: flat inside the zone, steep walls outside
- ±20 % on any weight keeps 7–10 of the top-10 zones
- 500 Monte Carlo weightings map where the result is stable

Code, data sources & notebooks:
github.com/MJFradi/mars-landing-site-suitability